Survey (Set 4)

When developers add new test cases into an existing test suite, do the developers look at the existing test cases and then reuse different attributes of existing test cases (such as variable naming, comments, test case naming, set of source code methods, overall test case organization/structure etc.) to derive the new testcases?

In order to answer the above question, we have designed this survey. The survey shows 20 pairs of unit test cases collected from open source projects. For each pair the participant is requested to indicate to what extent a developer *may/may not* have used a specific attribute to write the new test case.

Please answer all the following questions to complete the study. Thanks in advance.

* Indicates required question

Example 1: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
public void testInputStreamToByteArray() throws Exception {
    FileInputStream fin = new FileInputStream(m_testFile);
    try {
        byte[] out = IOUtils.toByteArray(fin);
        assertNotNull(out);
        assertTrue(""Not all bytes were read"", fin.available() == 0);
        assertTrue(""Wrong output size: out.length="" + out.length + ""!="" + FILE_SIZE, out.length == FILE_SIZE);
        assertEqualContent(out, m_testFile);
    } finally {
        fin.close();
    }
}
public void testInputStreamToBufferedInputStream() throws Exception {
    FileInputStream fin = new FileInputStream(m_testFile);
    try {
        InputStream in = IOUtils.toBufferedInputStream(fin);
        byte[] out = IOUtils.toByteArray(in);
        assertNotNull(out);
        assertTrue(""Not all bytes were read"", fin.available() == 0);
        assertTrue(""Wrong output size: out.length="" + out.length + ""!="" + FILE_SIZE, out.length == FILE_SIZE);
        assertEqualContent(out, m_testFile);
    } finally {
        fin.close();
    }
}
```

1. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

2. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

3. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

4. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

5. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

6. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

7. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

8. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

9. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 2: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testEncodeStringWithNull() throws Exception {
    final QuotedPrintableCodec qpcodec = new QuotedPrintableCodec();
    final String test = null;
    final String result = qpcodec.encode(test, "charset");
    assertNull(result, "Result should be null");
}
@Test
public void testTooShortByteArray() throws Exception {
    final QuotedPrintableCodec qpcodec = new QuotedPrintableCodec(true);
    assertNull(qpcodec.encode("AA"), "Result should be null.");
}
```

10. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

11. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

12. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

13. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

14. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

15. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

16. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

17. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

18. Please indicate if any other attributes were used (not listed in above)

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Untitled Section

Example 3: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testNoHeaderMap() throws Exception {
    try (final CSVParser parser =
            CSVParser.parse("a,b,c\n1,2,3\nx,y,z", CSVFormat.DEFAULT)) {
        Assert.assertNull(parser.getHeaderMap());
    }
}
@Test
public void testEmptyFileHeaderParsing() throws Exception {
    try (final CSVParser parser =
            CSVParser.parse("", CSVFormat.DEFAULT.withFirstRecordAsHeader())) {
        assertNull(parser.nextRecord());
        assertTrue(parser.getHeaderNames().isEmpty());
    }
}
```

19. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

20. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

21. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

22. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

23. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

24. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

25. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

26. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

27. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 4: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
public void testCarriageReturnLineFeedEndings() throws IOException {
    String code = "foo\r\nbaar,\r\nhello,world\r\n,kanu";
    CSVParser parser = new CSVParser(new StringReader(code));
    String[][] data = parser.getAllValues();
    assertEquals(4, data.length);
}
public void testCarriageReturnEndings() throws IOException {
    String code = "foo\rbaar,\rhello,world\r,kanu";
    CSVParser parser = new CSVParser(new StringReader(code));
    String[][] data = parser.getAllValues();
    assertEquals(4, data.length);
}
```

28. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

29. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

30. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

31. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

32. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

33. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

34. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

35. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

36. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 5: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"@Test
public void testPrint() throws IOException {
    final StringWriter sw = new StringWriter();
    try (final CSVPrinter printer =
            CSVFormat.DEFAULT.print(sw)) {
        printer.printRecord("a", "b\\c");
        assertEquals("a,b\\c" + recordSeparator, sw.toString());
    }
}
@Test
public void testPrintToFileWithDefaultCharset() throws IOException {
    File file = File.createTempFile(getClass().getName(), ".csv");
    try (final CSVPrinter printer =
            CSVFormat.DEFAULT.print(file, Charset.defaultCharset())) {
        printer.printRecord("a", "b\\c");
    }
    assertEquals("a,b\\c" + recordSeparator,
            FileUtils.readFileToString(file, Charset.defaultCharset()));
}"
```

37. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

38. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

39. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

40. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

41. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

42. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

43. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

44. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

45. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 6: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
@Test
public void testWrap_StringIntStringBooleanString() {
    //no changes test
    String input = "flammable/inflammable";
    String expected = "flammable/inflammable";
    assertEquals(expected, WordUtils.wrap(input, 30, "\n", false, "/"))
    // wrap on / and small width
    expected = "flammable\ninflammable";
    assertEquals(expected, WordUtils.wrap(input, 2, "\n", false, "/"));
    // wrap long words on / 1
    expected = "flammable\ninflammab\nle";
    assertEquals(expected, WordUtils.wrap(input, 9, "\n", true, "/"));
    // wrap long words on / 2
    expected = "flammable\ninflammable";
    assertEquals(expected, WordUtils.wrap(input, 15, "\n", true, "/"));
    // wrap long words on / 3
    input = "flammableinflammable";
    expected = "flammableinflam\nmable";
    assertEquals(expected, WordUtils.wrap(input, 15, "\n", true, "/"));
}
@Test
public void testLANG1292() throws Exception {
    // Prior to fix, this was throwing StringIndexOutOfBoundsException
    WordUtils.wrap("aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa " +
    "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa " +
    "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa", 70);
}
```

46. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

47. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

48. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

49. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

50. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

51. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

52. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

53. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

54. Please indicate if any other attributes were used (not listed in above)

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Untitled Section

Example 7: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void normalisesDocument() {
    String h = "<!doctype html>One<html>Two<head>Three<link
        ></head>Four<body>Five </body>Six </html>Seven ";
    Document doc = Jsoup.parse(h);
    assertEquals("<!doctype html><html><head><link /></head
        ><body>Five Six Seven  One Two Four Three</body
        ></html>", // is spaced OK if not newline & space
        stripped
    TextUtil.stripNewlines(doc.html()));
}
@Test
public void normalisesEmptyDocument() {
    Document doc = Jsoup.parse("");
    assertEquals("<html><head></head><body></body></html>",
        TextUtil.stripNewlines(doc.html()));
}
```

55. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

56. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

57. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

58. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

59. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

60. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

61. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

62. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

63. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 8: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testPlainQuoted() throws IOException {
    final StringWriter sw = new StringWriter();
    try (final CSVPrinter printer = new CSVPrinter(sw, CSVFormat.DEFAULT.withQuote('\'')) {
        printer.print("abc");
        assertEquals("abc", sw.toString());
    }
}
@Test
public void testEscapeBackslash3() throws IOException {
    StringWriter sw = new StringWriter();
    try (final CSVPrinter printer = new CSVPrinter(sw, CSVFormat.DEFAULT.withQuote(QUOTE_CH)) {
        printer.print("X\\\r");
    }
    assertEquals("'X\\\r'", sw.toString());
}
```

64. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

65. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

66. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

67. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

68. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

69. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

70. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

71. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

72. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 9: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
@Test
public void testSystemProperty() {
    final StringSubstitutor strSubst = StringSubstitutor
        .createInterpolator();
    final String spKey = "user.name";
    Assertions.assertEquals(System.getProperty(spKey), strSubst.replace
        ("${sys:" + spKey + "}"));
}
@Test
void testJavaScript() {
    Assertions.assertEquals("Hello World!", StringSubstitutor
        .createInterpolator().replace("${script:javascript:\"Hello
        World!\"}"));
    Assertions.assertEquals("7", StringSubstitutor.createInterpolator
        ().replace("${script:javascript:3 + 4}"));
}
```

73. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

74. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

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◯	◯	◯	◯	◯

75. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

76. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

77. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

78. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

79. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

80. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

81. Please indicate if any other attributes were used (not listed in above)

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Untitled Section

Example 10: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
@Test
public void testFileCleanerDirectory_NullStrategy() throws Exception {
    TestUtils.createFile(testFile, 100);
    assertTrue(testFile.exists());
    assertTrue(tempDirFile.exists());
    Object obj = new Object();
    assertEquals(0, theInstance.getTrackCount());
    theInstance.track(tempDirFile, obj, null);
    assertEquals(1, theInstance.getTrackCount());
    obj = null;
    waitUntilTrackCount();
    assertEquals(0, theInstance.getTrackCount());
    // not deleted, as dir not empty
    assertTrue(testFile.exists());
    // not deleted, as dir not empty
    assertTrue(testFile.getParentFile().exists());
}
@Test
public void testFileCleanerDirectoryFileSource() throws Exception {
    TestUtils.createFile(testFile, 100);
    assertTrue(testFile.exists());
    assertTrue(tempDirFile.exists());
    Object obj = new Object();
    assertEquals(0, theInstance.getTrackCount());
    theInstance.track(tempDirFile, obj);
    assertEquals(1, theInstance.getTrackCount());
    obj = null;
    waitUntilTrackCount();
    assertEquals(0, theInstance.getTrackCount());
    // not deleted, as dir not empty
    assertTrue(testFile.exists());
    // not deleted, as dir not empty
    assertTrue(testFile.getParentFile().exists());
}
```

82. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

83. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

84. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

85. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

86. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

87. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

88. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

89. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

90. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 11: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testExcelPrinter1() throws IOException {
    final StringWriter sw = new StringWriter();
    try (final CSVPrinter printer = new CSVPrinter(sw, CSVFormat.EXCEL)) {
        printer.printRecord("a", "b");
        assertEquals("a,b" + recordSeparator, sw.toString());
    }
}
@Test
public void testMongoDbCsvBasic() throws IOException {
    final StringWriter sw = new StringWriter();
    try (final CSVPrinter printer = new CSVPrinter(sw, CSVFormat.MONGODB_CSV)) {
        printer.printRecord("a", "b");
        assertEquals("a,b" + recordSeparator, sw.toString());
    }
}
```

91. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

92. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

93. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

94. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

95. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

96. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

97. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

98. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

99. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 12: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
@Test
public void testRelaxed() {
    String h = "<h1>Head</h1><td>One<td>Two</td>";
    String cleanHtml = Jsoup.clean(h, Whitelist.relaxed());
    assertEquals("<h1>Head</h1><table><tr><td>One</td><td>Two</td></tr></table>",
                 TextUtil.stripNewlines(cleanHtml));
}
@Test
public void testDropsUnknownTags() {
    String h = "<p><custom foo=true>Test</custom></p>";
    String cleanHtml = Jsoup.clean(h, Whitelist.relaxed());
    assertEquals("<p>Test</p>", cleanHtml);
}
```

100. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

101. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

102. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

103. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

104. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

105. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

106. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

107. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

108. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 13: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public final void testCompare_SurnamesCornerCase_MURPHY_NoSpace_NoMatch() {
    assertFalse(this.getStringEncoder().isEncodeEquals("Murphy", ""));
}
@Test
public final void testIsEncodeEquals_CornerCase_FirstNameNothing_ReturnsFalse() {
    assertFalse(this.getStringEncoder().isEncodeEquals("", "test"));
}
```

109. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

110. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

111. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

112. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

113. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

114. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

115. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

116. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

117. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 14: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testGetAsLong() {
    assertThrows(UncheckedIOException.class, () -> Uncheck.getAsLong(() -> {
        throw new IOException();
    }));
    assertThrows(UncheckedIOException.class,
       () -> Uncheck.getAsLong(TestConstants.THROWING_IO_LONG_SUPPLIER));
    assertEquals(1L, Uncheck.getAsLong(() -> TestUtils.compareAndSetThrowsIO(atomicLong, 1L)));
    assertEquals(1L, atomicLong.get());
}
@Test
public void testGetAsBoolean() {
    assertThrows(UncheckedIOException.class, () -> Uncheck.getAsBoolean(() -> {
        throw new IOException();
    }));
    assertThrows(UncheckedIOException.class,
        () -> Uncheck.getAsBoolean(TestConstants.THROWING_IO_BOOLEAN_SUPPLIER));
    assertTrue(Uncheck.getAsBoolean(() -> TestUtils.compareAndSetThrowsIO(atomicBoolean, true)));
    assertTrue(atomicBoolean.get());
}
```

118. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

119. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

120. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

121. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

122. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

123. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

124. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

125. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

126. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 15: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"@Test
public void parentChildStar() {
    String h = "<div id=1><p>Hello<p><b>there</b></p></div><div id
        =2><span>Hi</span></div>";
    Document doc = Jsoup.parse(h);
    Elements divChilds = doc.select(""div > *"");
    assertEquals(3, divChilds.size());
    assertEquals(""p"", divChilds.get(0).tagName());
    assertEquals(""p"", divChilds.get(1).tagName());
    assertEquals(""span"", divChilds.get(2).tagName());
}
@Test
public void adjacentSiblingsWithId() {
    String h = "<ol><li id=1>One<li id=2>Two<li id=3>Three</ol>";
    Document doc = Jsoup.parse(h);
    Elements sibs = doc.select(""li#1 + li#2"");
    assertEquals(1, sibs.size());
    assertEquals(""Two"", sibs.get(0).text());
}"
```

127. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

128. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

129. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

130. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

131. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

132. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

133. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

134. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

135. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 16: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"@Test
public void unexpectedStringWhileDecodingTest() {
    final String toDecode = ""00XX"";
    assertThatThrownBy(() -> {
        final AlphabetConverter ac = createJavadocExample();
        ac.decode(toDecode);
    }).isInstanceOf(UnsupportedEncodingException.class).
       hasMessage(""Unexpected string without decoding (XX) in "" + toDecode);
}
@Test
public void unexpectedEndWhileDecodingTest() {
    final String toDecode = ""00d01d0"";
    assertThatThrownBy(() -> {
        final AlphabetConverter ac = createJavadocExample();
        ac.decode(toDecode);
    }).isInstanceOf(UnsupportedEncodingException.class).
        hasMessage(""Unexpected end of string while decoding "" + toDecode);
}
```

136. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

137. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

138. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

139. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

140. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

141. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

142. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

143. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

144. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 17: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"@Test
public void testPrintNullValues() throws IOException {
    final StringWriter sw = new StringWriter();
    try (final CSVPrinter printer = new CSVPrinter(sw, CSVFormat.DEFAULT)) {
        printer.printRecord(""a"", null, ""b"");
        assertEquals(""a,,b"" + recordSeparator, sw.toString());
    }
}
@Test
public void testNotFlushable() throws IOException {
    final Appendable out = new StringBuilder();
    try (final CSVPrinter printer = new CSVPrinter(out, CSVFormat.DEFAULT)) {
        printer.printRecord(""a"", ""b"", ""c"");
        assertEquals(""a,b,c"" + recordSeparator, out.toString());
        printer.flush();
    }
}"
```

145. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

146. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

147. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

148. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

149. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

150. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

151. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

152. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

153. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 18: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"@Test
public void testGoogleSearchIpod() {
    String h = loadFile(""/htmltests/google-ipod.html"");
    Document doc = Jsoup.parse(h, "http://www.google.com/search?hl=en&q=ipod&aq
        =f&oq=&aqi=g10");
    assertEquals(""ipod - Google Search"", doc.getTitle());
    Elements results = doc.select(""h3.r > a"");
    assertEquals(12, results.size());
    assertEquals("http://news.google.com/news?hl=en&q=ipod&um=1&ie=UTF-8&ei
        =uYlKS4SbBoGg6gPf-5XXCw&sa=X&oi=news_group&ct=title&resnum=1&ved
        =0CCIQsQQwAA", results.get(0).attr("href"));
    assertEquals("http://www.apple.com/itunes/", results.get(1).attr("href"));
}
@Test
public void testYahooJp() {
    String h = loadFile(""/htmltests/yahoo-jp.html"");
    // http charset is utf-8.
    Document doc = Jsoup.parse(h, "http://www.yahoo.co.jp/index.html");
    assertEquals(""Yahoo! JAPAN"", doc.getTitle());
    Element a = doc.select(""a[href=t/2322m2]"").first();
    assertEquals("http://www.yahoo.co.jp/_ylh
        =X3oDMTB0NWxnaGxsBF9TAzIwNzcyOTYyNjUEdGlkAzEyBHRtcGwDZ2Ex/t/2322m2", //
        session put into <base>
    a.attr(""abs:href""));
    assertEquals(""全国、人気の駅ランキング"", a.text());
}"
```

154. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

155. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

156. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

157. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

158. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

159. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

160. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

161. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

162. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 19: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testNull() {
    Assertions.assertNull(
        ResourceBundleStringLookup.INSTANCE.lookup(null));
}
@Test
public void testBadName() {
    assertThrows(IllegalArgumentException.class, () -> {
        ResourceBundleStringLookup.INSTANCE.lookup(""BAD_RESOURCE_BUNDLE_NAME:KEY"");
    });
}
```

163. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

164. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

165. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

166. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

167. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

168. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

169. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

170. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

171. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 20: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
@Test
public void testAddNewText() {
    Document doc = Jsoup.parse("<div id=1><p>Hello</p></div>");
    Element div = doc.getElementById(""1"");
    div.appendText(" there & now >");
    assertEquals("<p>Hello</p> there &amp; now &gt;", div.html());
}
@Test
public void testPrependText() {
    Document doc = Jsoup.parse("<div id=1><p>Hello</p></div>");
    Element div = doc.getElementById(""1"");
    div.prependText("there & now > ");
    assertEquals("there &amp; now &gt; <p>Hello</p>", div.html());
}
```

172. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

173. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

174. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

175. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

176. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

177. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

178. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

179. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

180. Please indicate if any other attributes were used (not listed in above)

_____

Demographic Information

181. What is you current post * Dropdown

Mark only one oval.

- PhD Student
- Masters Student
- Undergraduate Student
- Software Developer
- Others

182. How many years of programming experience you have? Dropdown

Mark only one oval.

- 0-1 year
- 1-3 years
- 3-5 years
- more than 5 years

183. Do you have experience in writing unit test cases?

Mark only one oval.

- Yes
- No
- Maybe

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